Lab Report: Comparison of HPGe and NaI(Tl) Detectors in Gamma-Ray Spectroscopy

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1 Introduction

The main goal of this report is to differentiate between the High Purity Germanium detectors - HPGe and the Sodium Iodide Scintillation detectors - NaI(Tl) in gamma-ray spectroscopy. Subsequently, follows the calculation of mean free path length of 662 keV photons in water using a HPGe detector.

2 Materials and Methods

2.1 HPGe Detectors

2.1.1 Operating Principles

1. Gamma rays interact with the Germanium crystal via photoelectric effect, Compton scattering, or pair production.
2. The interaction produces electron-hole pairs in the crystal.
3. An applied electric field sweeps the charges to electrodes, creating a current pulse.
4. The pulse is amplified and analyzed to measure the gamma-ray energy with high precision.

2.1.2 Main Characteristics

- Superior energy resolution, allowing for precise detection and identification of gamma-ray energies.
- Effective over a broad energy range (hundreds of eV to several MeV), making them suitable for a plethora of different applications.
- Require cryogenic operations, in order to maintain performance and are high-cost.

2.2 NaI(Tl) Detectors

2.2.1 Operating Principles

1. Gamma rays interact with the sodium iodide crystal, producing flashes of visible light.
2. The light is collected by a photomultiplier tube (PMT) or photodiode.
3. Electronics process the signal to form a gamma-ray spectrum.
4. The PMT converts the light into an electrical signal proportional to gamma-ray energy.

2.2.2 Main Characteristics

- High detection efficiency due to high light yield.
- Versatile, durable, and cost-effective.
- Less precise than HPGe detectors.

2.3 Summary

- HPGe detectors are preferred for high energy resolution and detailed spectral imaging (e.g., scientific research, nuclear security).
- NaI(Tl) detectors are suitable for applications with limited funding or lower precision requirements (e.g., environmental monitoring, medical imaging).

3 Energy Spectrum Calibration

3.1 HPGe Energy Calibration

A linear equation was computed based on the points:

$$(\mathrm{Ch}_1, \mathrm{En}_1) = (1776, 662)$$
$$(\mathrm{Ch}_2, \mathrm{En}_2) = (3940, 1461)$$

shown here:

$$En = a \cdot \mathrm{Ch} + B \tag{1}$$

where:

$$a = (\mathrm{En}_2 - \mathrm{En}_1)/(\mathrm{Ch}_2 - \mathrm{Ch}_1) \tag{2}$$

and

$$b = \mathrm{En}_2 - a \cdot \mathrm{Ch}_2 \tag{3}$$

The equation was used to transform the x-axis values from channels to energies.

3.2 NaI(Tl) Energy Calibration

A polynomial fit is performed, and a 2nd-degree polynomial model is applied to transform the x-axis values. The R-squared values of the tested fits are:

- Degree 1: $R^2 = 0.9483$
- Degree 2: $R^2 = 0.9998$
- Degree 3: $R^2 = 0.9999$
- Degree 4: $R^2 = 1.0000$
- Degree 5: $R^2 = 1.0000$

While higher Degree models have slightly improved R-squared values i choose the 2nd degree polynomial model to avoid over-fitting and to take advantage of its simplicity.

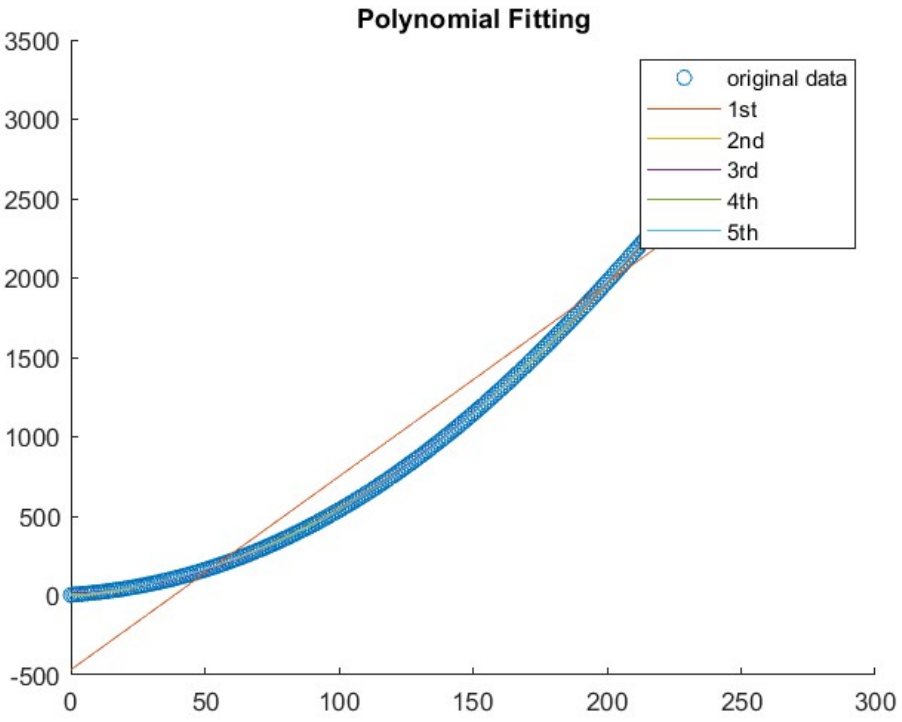
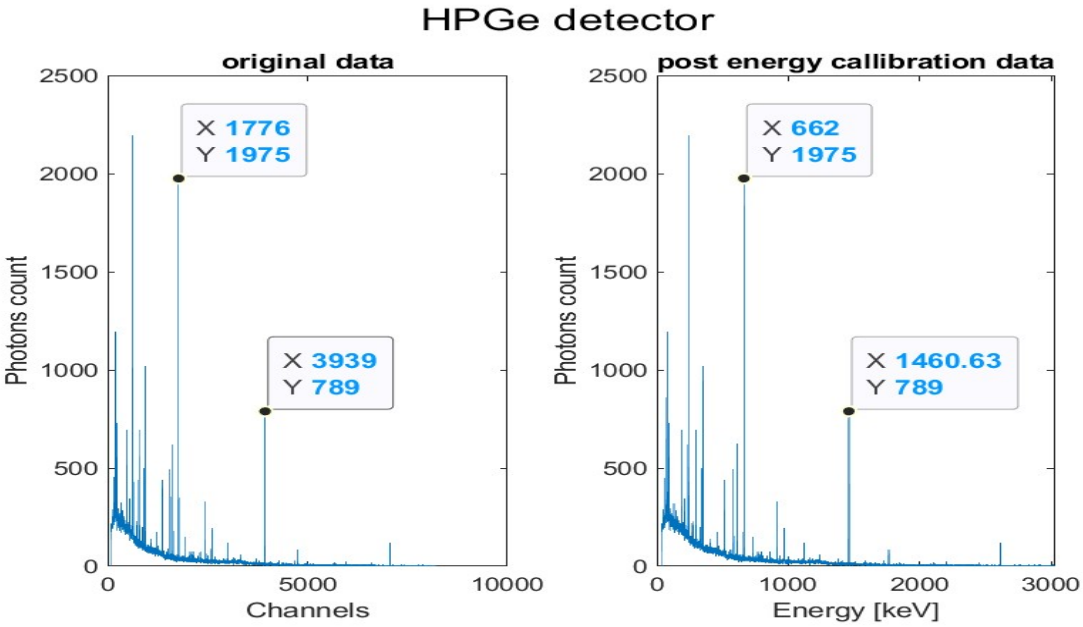
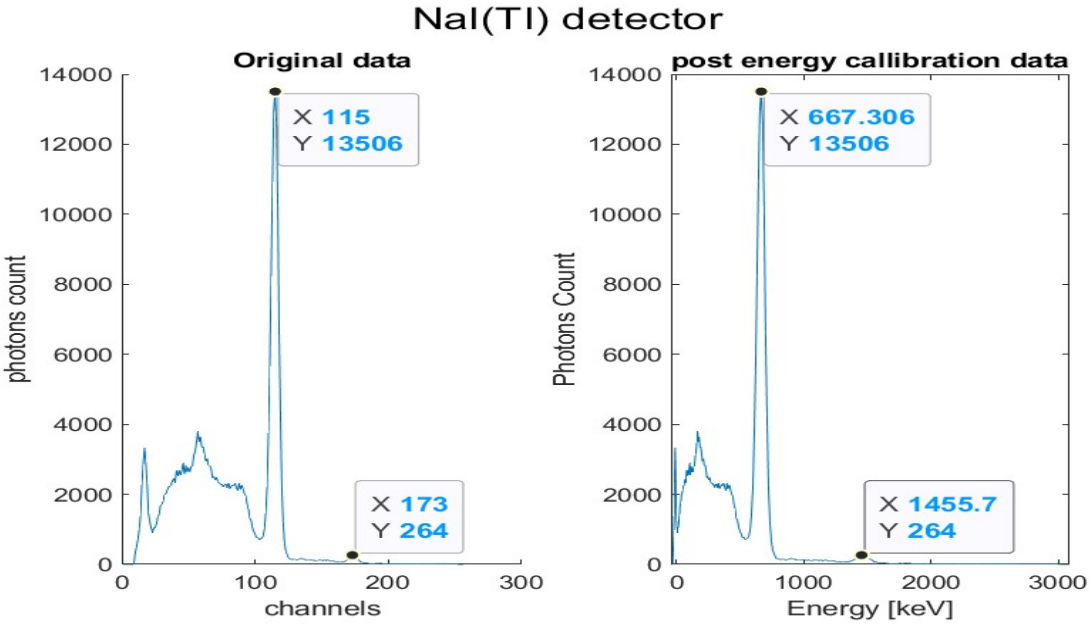


Figure 1: Polynomial fitting using (Ch, En) pairs

3.3 Channel-Photon Counts vs Energy-Photon Counts

Graphs of the data for each detector are presented, both before and after applying the energy calibration.



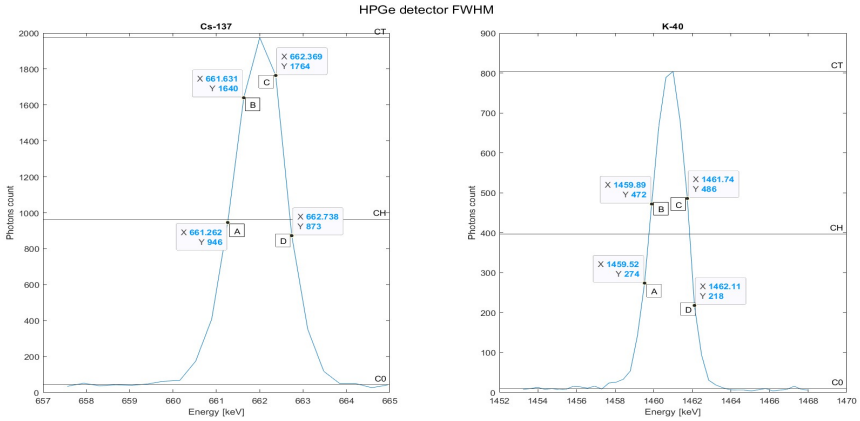


4 FWHM (Full Width Half Maximum) Estimation

4.1 HPGe Detector

Using the method of "Interpolation between channels":
where C_T = full height of peak

- Cs-137 peak (662 keV): FWHM $\approx$ 1.9764 keV ($\frac{FWHM}{C_T} \approx 0.1\%$).
- K-40 peak (1461 keV): FWHM $\approx$ 1.9308 keV ($\frac{FWHM}{C_T} \approx 0.24\%$).



Using the "area under peak" formula:

$$FWHM = 0.939 \cdot \frac{A}{C_T - C_O} \tag{4}$$

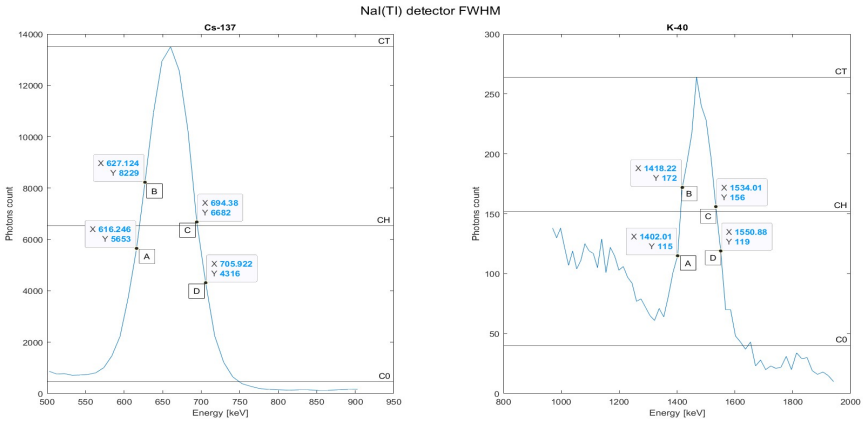
where A is the area of the peak and the denominator is the full height of the peak (C_T) corrected for underlying background (C_O).

- Cs-137 peak (662 keV): $C_T = 1975$ and $C_O = 45$
  FWHM $\approx$ 1.9336 keV ($\frac{FWHM}{C_T} \approx 0.1\%$).
- K-40 peak (1461 keV): $C_T = 804$ and $C_O = 10$
  FWHM $\approx$ 2.1182 keV ($\frac{FWHM}{C_T} \approx 0.15\%$).

4.2 NaI(Tl) Detector

Using the method of "Interpolation between channels":

- Cs-137 peak (662 keV): FWHM $\approx$ 77.8954 keV ($\frac{FWHM}{C_T} \approx 0.58\%$).
- K-40 peak (1461 keV): FWHM $\approx$ 131.4590 keV ($\frac{FWHM}{C_T} \approx 49.80\%$).



Using the "area under peak" formula:

$$FWHM = 0.939 \cdot \frac{A}{C_T - C_O} \tag{5}$$

where A is the area of the peak and the denominator is the full height of the peak (C_T) corrected for underlying background (C_O).

- Cs-137 peak (662 keV): $C_T = 13506$ and $C_O = 450$
  FWHM $\approx$ 83.2497 keV ($\frac{FWHM}{C_T} \approx 0.62\%$).
- K-40 peak (1461 keV): $C_T = 264$ and $C_O = 40$
  FWHM $\approx$ 149.2983 keV ($\frac{FWHM}{C_T} \approx 56.55\%$).

4.3 Observations and remarks:

- As shown above and in the MATLAB script, the "Area under peak" method generally agrees with the "Interpolation between channels", as long as the bounds of the calculated area are correctly identified.

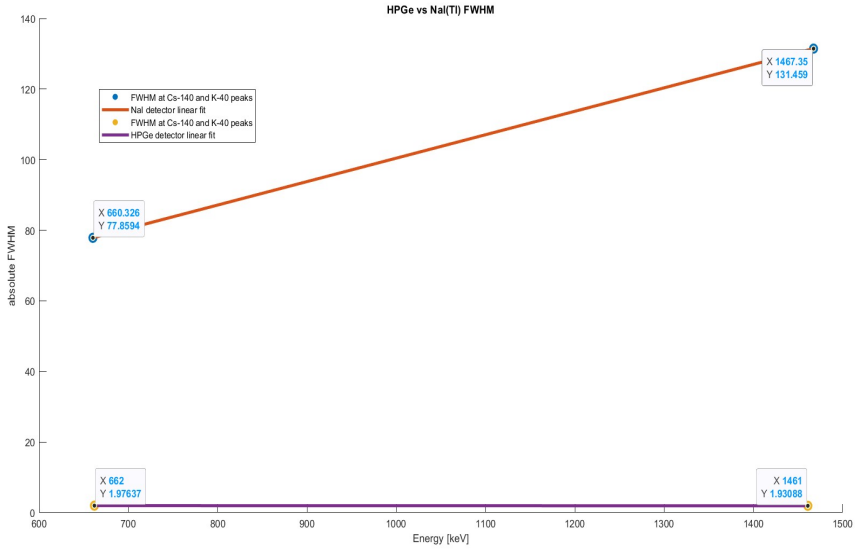
- It is clear that the FWHM of the HPGe detector is significantly lower than the FWHM of the NaI(Tl) detector. At the Cs-137 peak the FWHM of the HPGe detector is about 6 times lower than the one of NaI(Tl) detector and, at the K-40 peak it is about 65 times lower. Showing why the resolution of the HPGe detector is so much higher than that of the NaI(Tl).

- Furthermore, it seems that the energies close to the K-40 peak are barely detected by the NaI(Tl) detector, resulting in a "weak" peak. This shows the known attribute of this kind of detectors that they are not as effective in high energy detection as they are in lower energy levels.

Summary: The FWHM of the HPGe detector is significantly lower than that of the NaI(Tl) detector, demonstrating its superior resolution.

4.4 FWHM-Energy relation

Below is shown a figure demonstrating each detector's FWHM-Energy relation based on the FWHM values we calculated at the peaks of Cs-137 and K-40. The points of calibration are minimal (only two), but we can still make some observations.



Each line is described by the equation:

$$FWHM = a \cdot Energy + b \tag{6}$$

where $a = \frac{FWHM_2 - FWHM_1}{E_2 - E_1}$

- In the HPGe detector the FWHM does not change significantly for different values of energy: FWHM = 1.97 keV for E = 662keV and FWHM = 1.93 keV for E = 1461keV.

$$a = -5.7 \cdot 10^{-5} = -0.000057 \text{ and } b = 1.97673$$

- In the NaI(Tl) there is a radical increment of the FWHM while increasing the corresponding Energy: FWHM = 77.85 keV for E = 660keV and FWHM = 131.46 keV for E = 1467keV.

$$a = 0.0664 \text{ and } b = 77.8594$$

5 Calculation of the Mean Free Path Length of 662 keV Photons in Water

5.1 Theoretical Introduction

The photon attenuation law describes the rate with which the gamma photons decrease, while passing through a material, due to different interactions (e.g. photoelectric absorption, Compton effect, pair production)

$$I(x) = I_0 \cdot e^{-\mu x} \tag{7}$$

where:

- $I(x)$: Intensity after traveling distance x (cm).
- I_0: Initial intensity of the photons.
- μ: Linear attenuation coefficient (cm^{-1}).

5.1.1 Key definitions:

- μ: Linear attenuation coefficient: expresses the probability of photon interaction inside the material and depends on the material and photon energy.
- μ/ρ: Mass attenuation coefficient: normalizes μ by introducing the density(ρ) of the material, is often used to compare materials. (cm^2/g).
- Relationship between the above: $\mu = (\mu/\rho) \cdot \rho$

5.2 Cross-Sections and Mean Free Path

Microscopic Cross-Section (σ): the probability of an interaction occurring for a simple atom or nucleus. (cm^2 or barns). It depends on:

- Type of interaction (e.g., Compton effect, scattering, photoelectric absorption, etc.)
- Energy of the particle
- Material in which the interaction occurs

Macroscopic Cross-Section (Σ): the total probability of any interaction occurring considering all atoms or nuclei (units: cm^{-1}). It does not depend on the type of interaction as it expresses the summed probability of all interactions.

$$\Sigma_t = \sum_i \sigma_i N_i$$

where Σ_t is the macroscopic cross-section and N_i is the number of atoms or nuclei per unit volume.

Mean Free Path of Photons (λ): the average distance that photons travel through a material before taking part in an interaction. It is directly related to the material's linear attenuation coefficient (μ) or macroscopic cross-section (Σ_t).

$$\lambda = \frac{1}{\mu} \quad \text{or} \quad \lambda = \frac{1}{\Sigma_t}$$

where:

- λ is the mean free path (cm)
- μ is the linear attenuation coefficient (cm^{-1})
- Σ_t is the macroscopic cross-section (cm^{-1})

5.3 Experimental Results

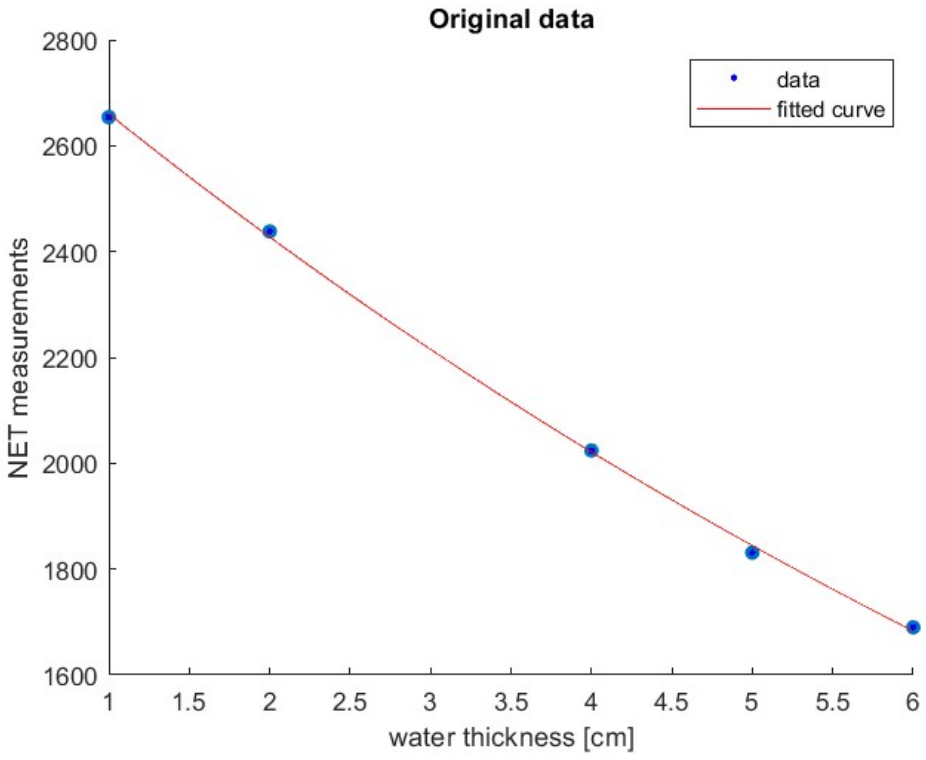
Experiment summary: calculation of the mean free path length of 662 keV photons in water, using a Cs-137 radiation source and an HPGe detector. The experiment, resulted in 5 NET measurements:

$$(x_1, NET_1) = (1, 2654)$$
$$(x_2, NET_2) = (2, 2438)$$
$$(x_3, NET_3) = (4, 2024)$$
$$(x_4, NET_4) = (5, 1831)$$
$$(x_5, NET_5) = (6, 1690)$$

where x_i is the water thickness for each iteration i and NET_i is the count of photons detected for each iteration i.

I used two different approaches to calculate μ and λ

1. Exponential fit (MATLAB): $\mu = 0.09158\,\mathrm{cm}^{-1}$, $\lambda = 10.9114\,\mathrm{cm}$.



2. Ratio method: $\mu = 0.0909\,\mathrm{cm}^{-1}$, $\lambda = 11.0022\,\mathrm{cm}$.
   (Using the ratio of all the different combinations of the net measurements and calculating the mean value of those:

$$I(x) = I_0 \cdot e^{-\mu x}$$

   and

$$I(y) = I_0 \cdot e^{-\mu y}$$

   then

$$\mu_{xy} = -\frac{log(\frac{I(y)}{I(x)})}{y - x}$$

$$\implies \mu_{ij} = -\frac{log(\frac{NET_j}{NET_i})}{x_j - x_i}$$

   giving us the following:

$$\mu_{12} = 0.0849$$

$$\mu_{13} = 0.0903$$

$$\mu_{14} = 0.0928$$

$$\mu_{15} = 0.0903$$

$$\mu_{23} = 0.0931$$

$$\mu_{24} = 0.0954$$

$$\mu_{25} = 0.0916$$

$$\mu_{34} = 0.1002$$

$$\mu_{35} = 0.0902$$

$$\mu_{45} = 0.0801$$

$$\implies mean(\mu_{ij}) = \frac{\sum \mu_{ij}}{\binom{max(i)}{2}} = 0.0909\ \mathrm{cm}^{-1}$$

   MATLAB software is not necessary for this method.
   *keep in mind that this is not a verified method but just an idea so it may be invalid.

Comparison with literature values:

- Values calculated from book table (2.10.1):
  $\mu = 0.0862\,\mathrm{cm}^{-1}$, $\lambda = 11.6023\,\mathrm{cm}$.
  supposed linear relation between μ/ρ and energy and performed linear interpolation:
  regarding the aforementioned table: if $\mu(E)$ is the attenuation coefficient of water for different energies E in MeV:
  $\mu(0.6) = 0.0896$ and $\mu(0.8) = 0.0786$ then using the linear equation:

$$\mu(E) = a \cdot E + b \tag{8}$$

  where

$$a = \frac{\mu(0.8) - \mu(0.6)}{0.8 - 0.6} = -0.055$$

  and

$$b = \mu(0.6) - a \cdot 0.6 = 0.1226$$

  we end up with:

$$\mu(0.662) = a \cdot 0.662 + 0.1226$$
$$\implies \mu(0.662) = 0.0896\mathrm{cm}^{-1}$$

Deviations:

- 1st approach: err $= -0.00702\mathrm{cm}^{-1}$ ($\approx 7.73\%$).
- 2nd approach: err $= -0.0077\mathrm{cm}^{-1}$ ($\approx 7.81\%$).

Possible sources of deviation include water impurities, calibration errors, and temperature variations.
*negative errors mean that the experimental method resulted in lower μ than expected in this instance.